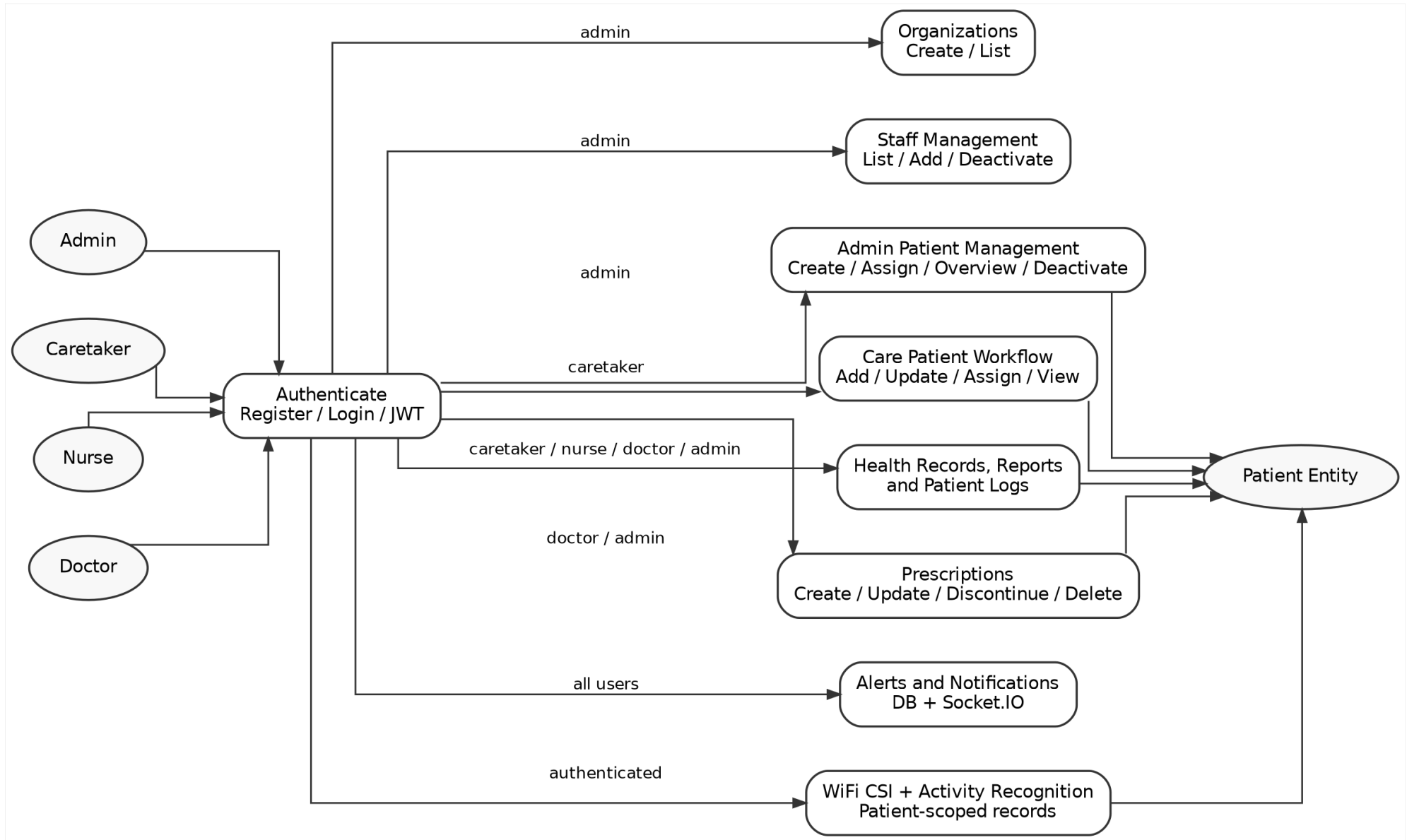
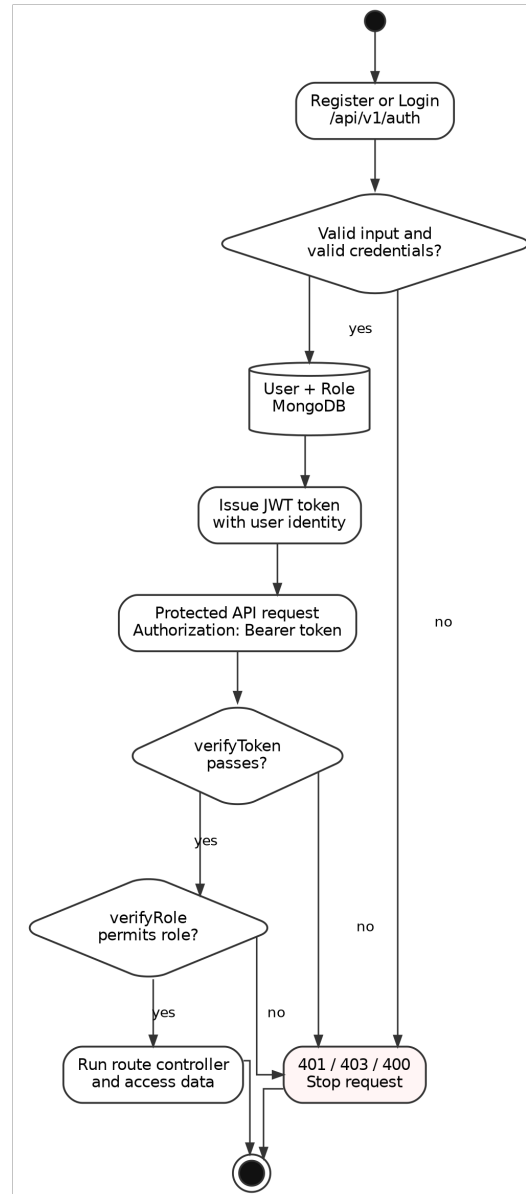


Overall Backend User Entity Access Map



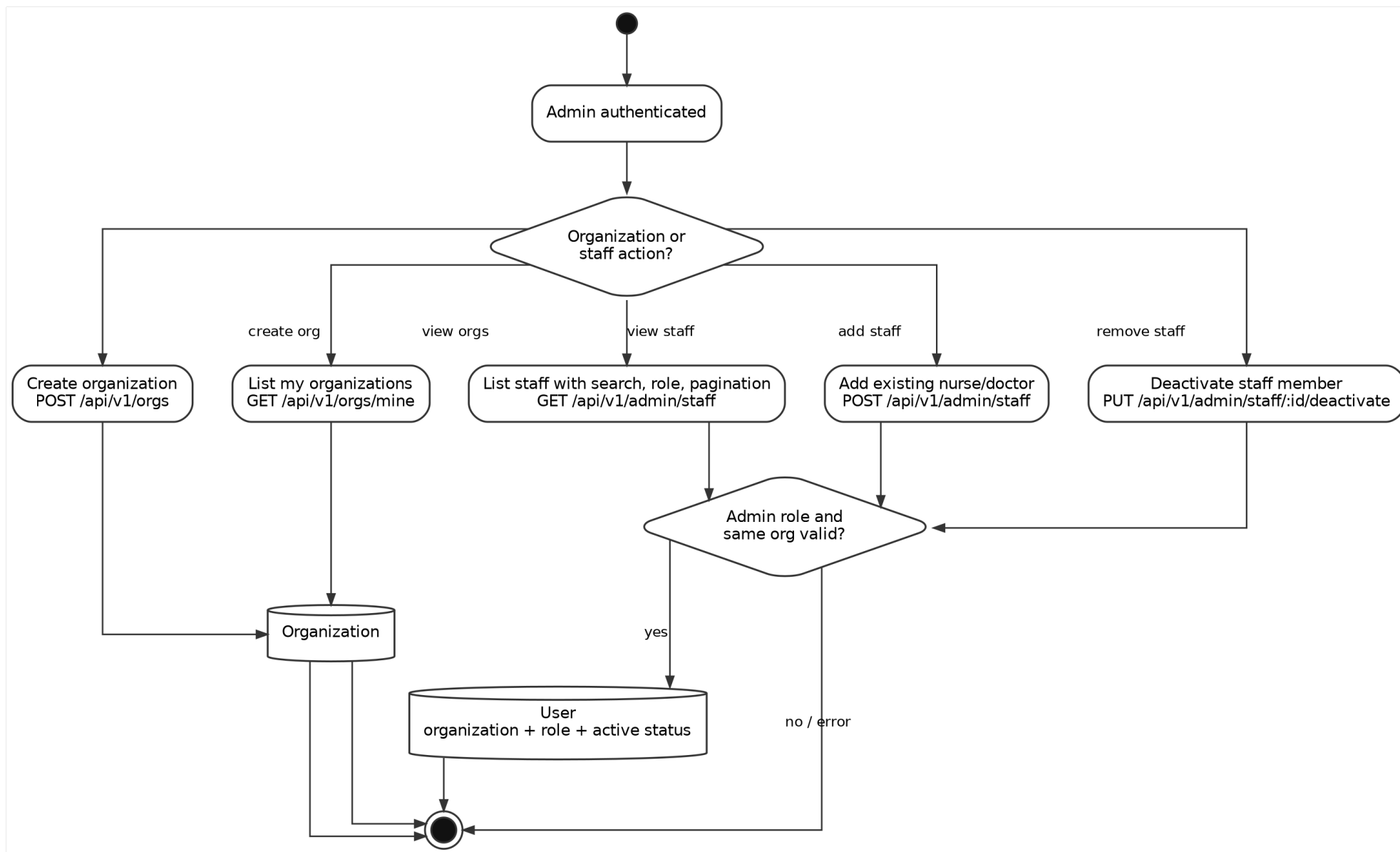
This UML use-case style overview shows how the backend separates access by user entity. All users enter through authentication, while Admin manages organizations, staff, patients, resources, tasks and support tickets; Caretaker manages assigned patients and daily care reporting; Nurse manages assigned patient care records and logs; Doctor manages clinical review and prescriptions; and the Patient entity receives records, logs, prescriptions and sensor-linked monitoring data.

Authentication and RBAC Workflow - All Users



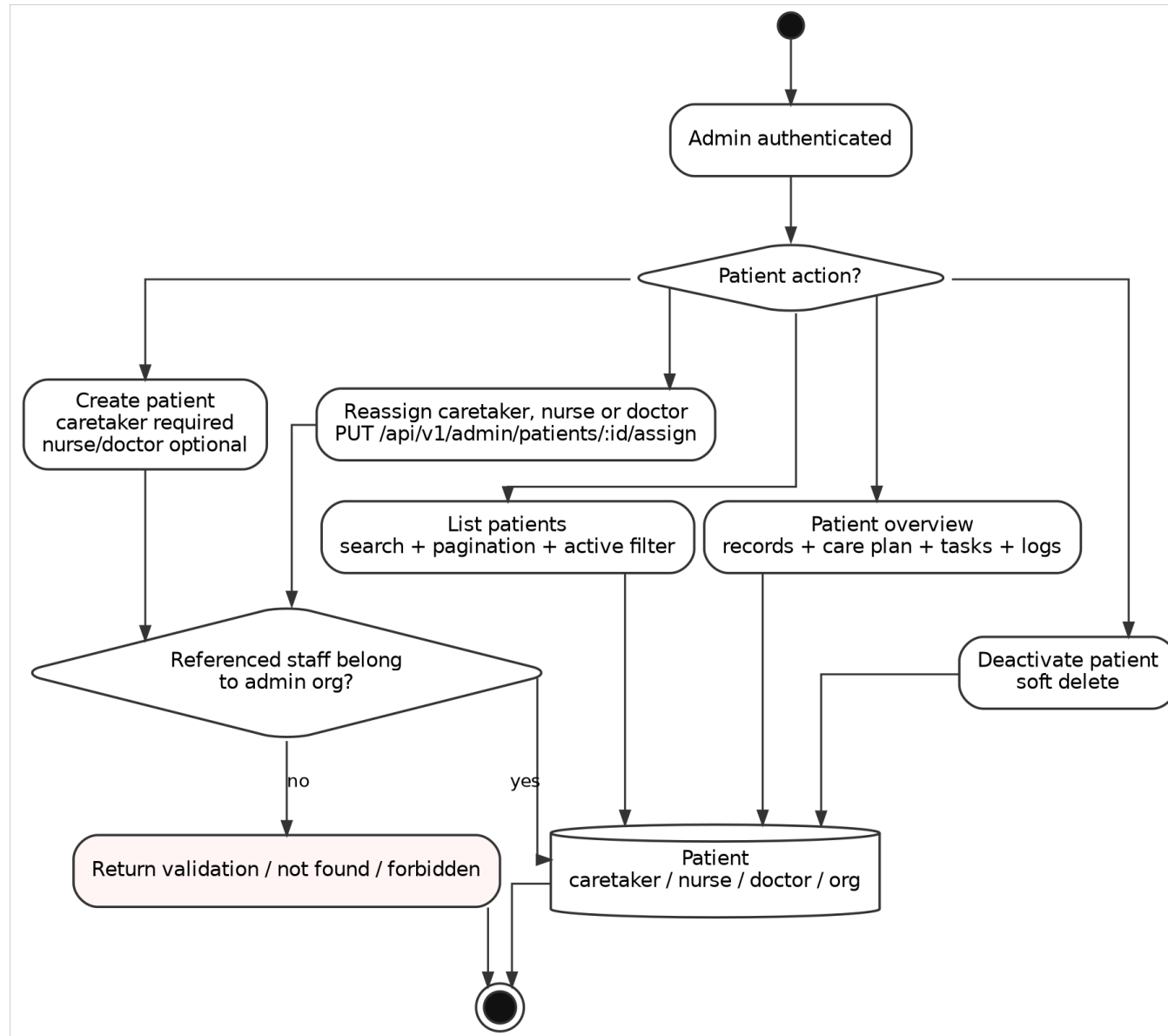
This activity workflow applies before protected backend actions. A user registers or logs in, validation and password checks are completed, a JWT is issued, then protected requests pass through verifyToken and role checks before the controller is allowed to run. If the token is missing/invalid or the role is not permitted, the backend stops the request with an unauthorized or forbidden response.

Admin Workflow - Organization and Staff Management



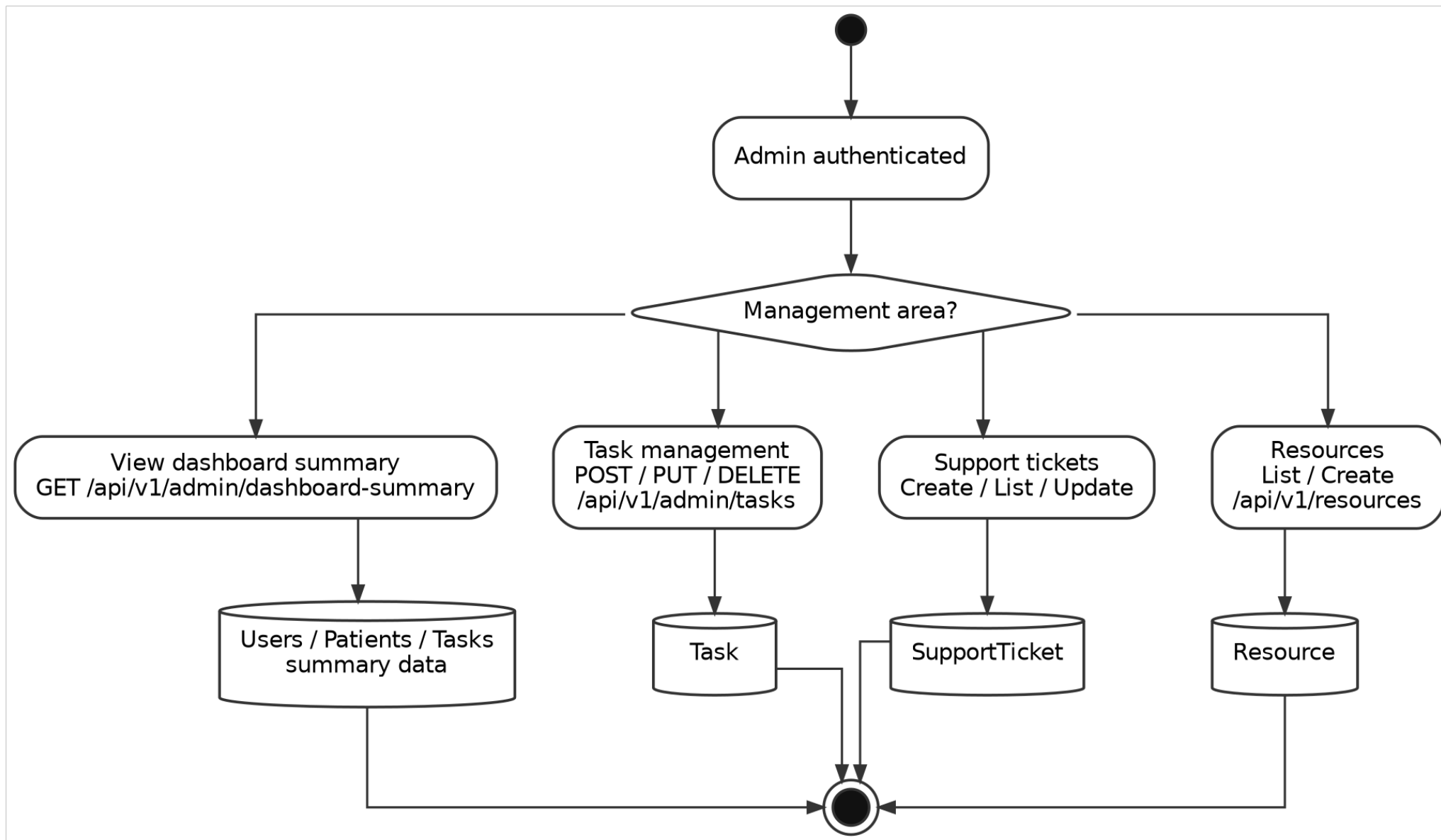
This admin workflow covers organization and staff setup for the organization-based mode. After login, the admin creates or views their organizations, then manages nurses and doctors through staff listing, adding existing users into the organization, and soft deactivating staff when they should no longer belong to the active care team.

Admin Workflow - Patient Assignment and Oversight



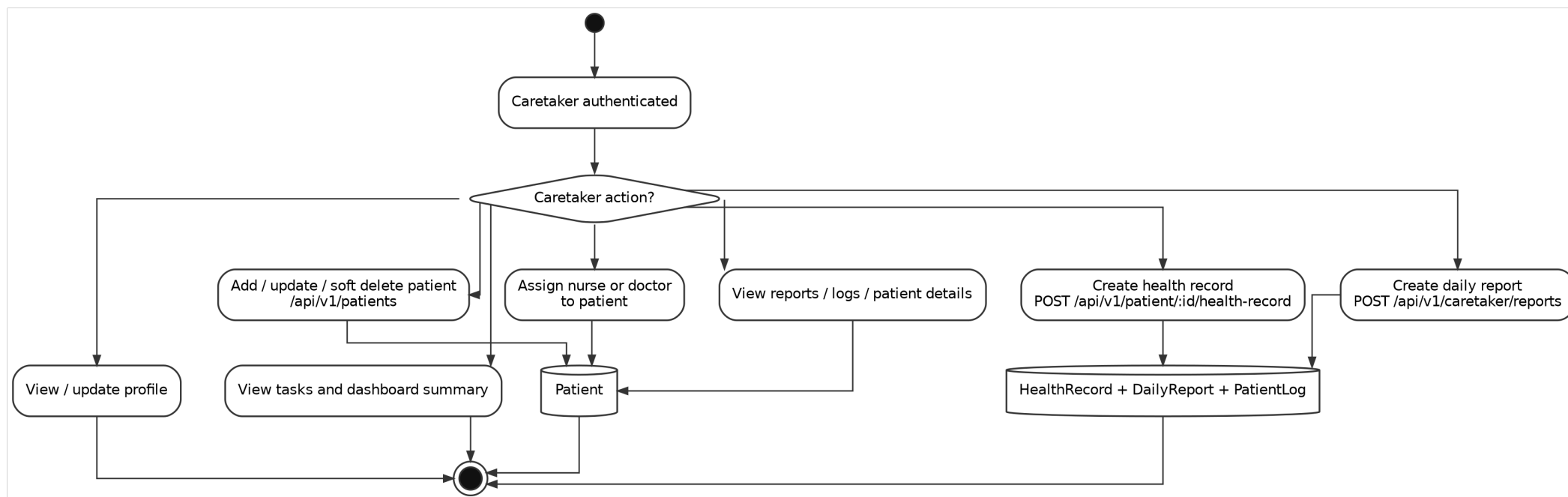
This admin patient workflow shows how the backend supports organization-level patient management. The admin can create a patient under a caretaker, optionally attach a nurse and doctor, reassign care roles later, view the full patient overview containing records, care plan, tasks and logs, or deactivate the patient using a soft-delete style action.

Admin Workflow - Tasks, Support Tickets, Dashboard and Resources



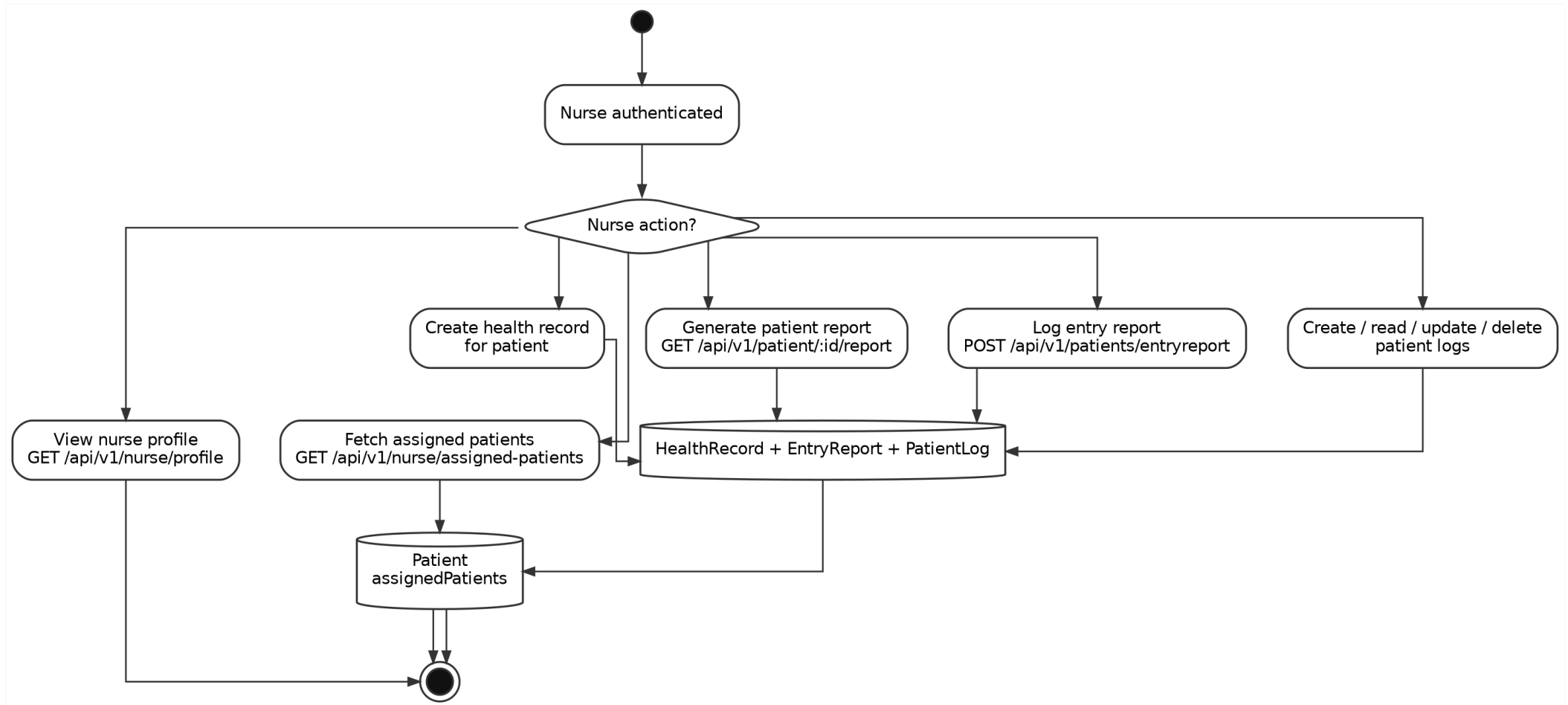
This admin workflow captures the supporting management features around the main care system. Admin users can review dashboard summary data, create/update/delete tasks, manage support tickets, and maintain admin-only resources, giving the next team a clear view of the management functions that sit beside the patient and staff workflows.

Caretaker Workflow - Patient Care and Daily Reporting



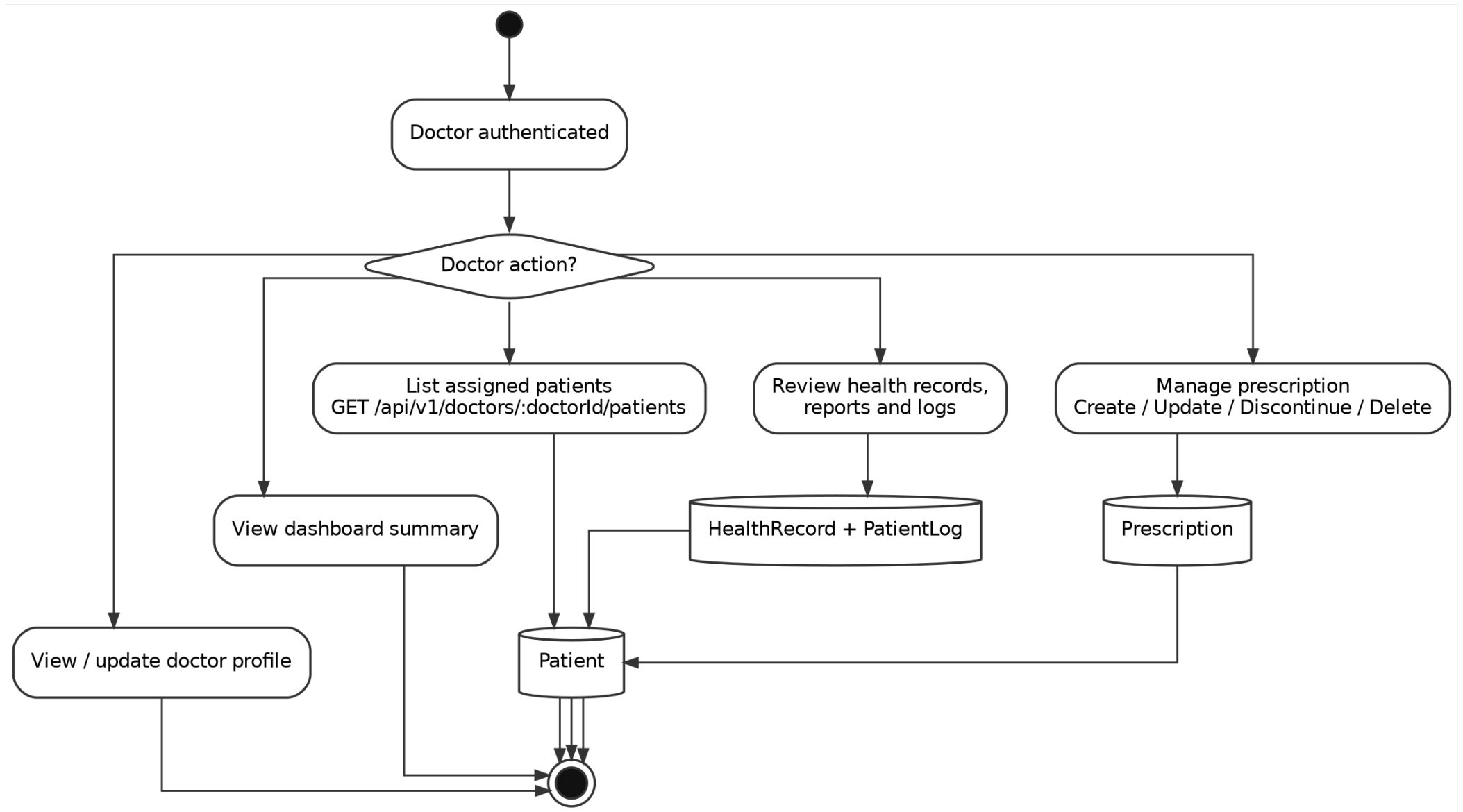
This caretaker workflow covers the operational patient-care path. A caretaker logs in, views profile, tasks and dashboard summary, creates or updates assigned patients, assigns a nurse or doctor when required, adds health records, submits daily reports, and views patient-specific reports and logs that support continuous care tracking.

Nurse Workflow - Assigned Patient Monitoring



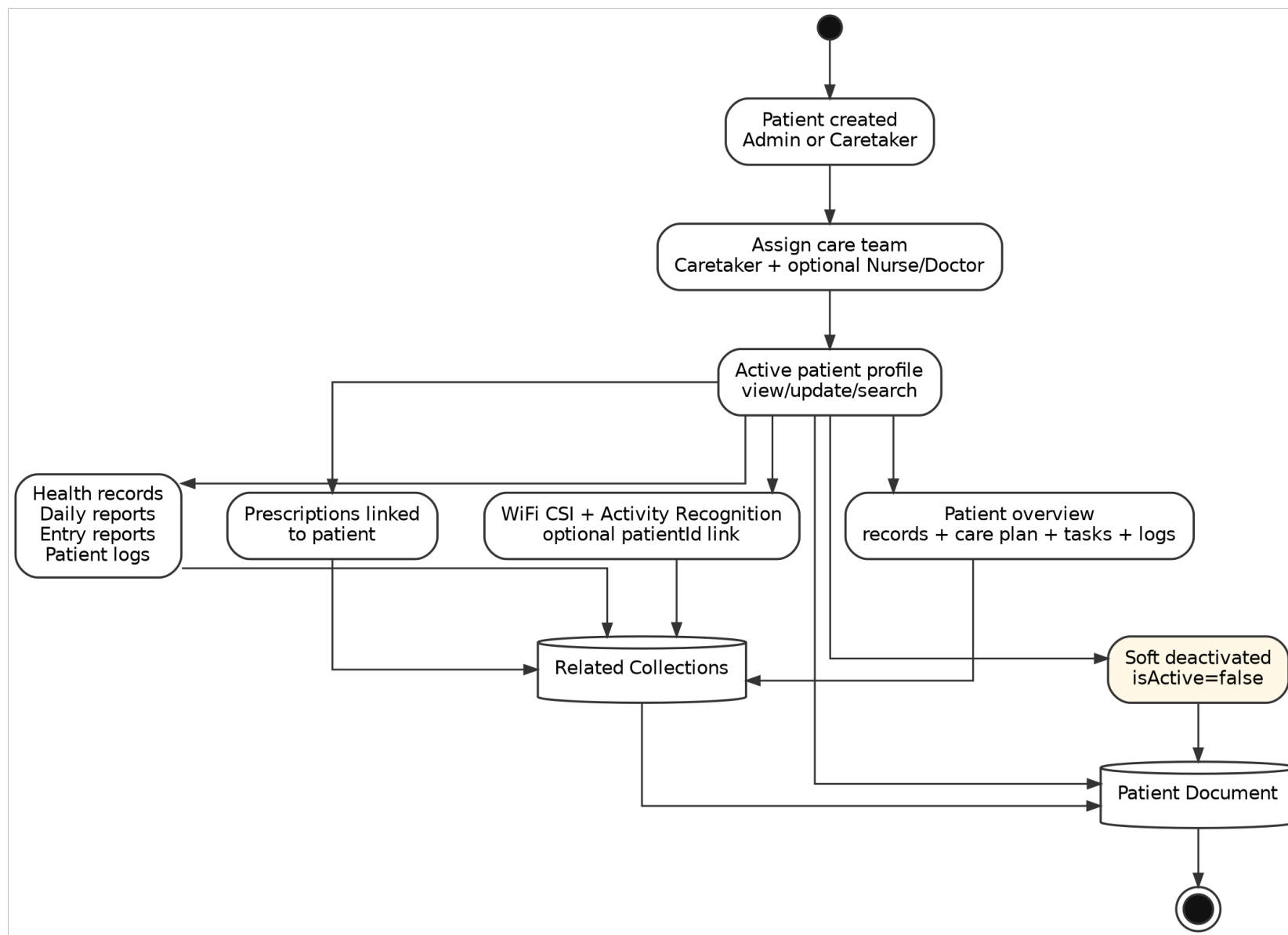
This nurse workflow shows how nurses work through their assigned patients. After authentication, a nurse can view their profile, fetch assigned patients, add patient health records, generate patient reports, create entry reports, and contribute to patient logs so clinical observations are connected back to the correct patient record.

Doctor Workflow - Patient Review and Prescriptions



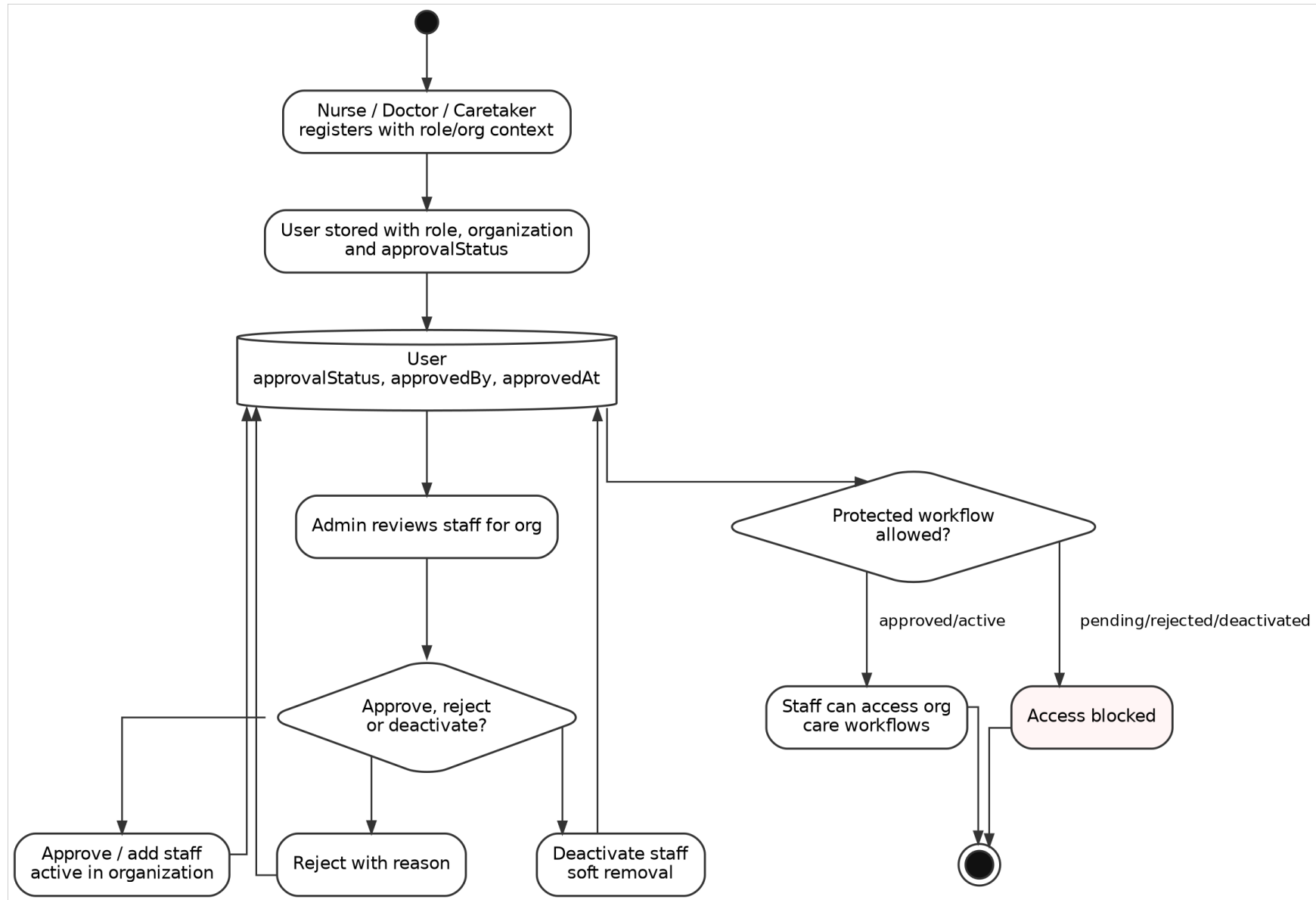
This doctor workflow focuses on clinical review. A doctor can maintain their profile, view dashboard summary, list patients assigned to them, review health records, reports and patient logs, and manage prescriptions through create, update, discontinue and delete operations where the backend allows doctor or admin access.

Patient Entity Workflow - Lifecycle and Linked Data



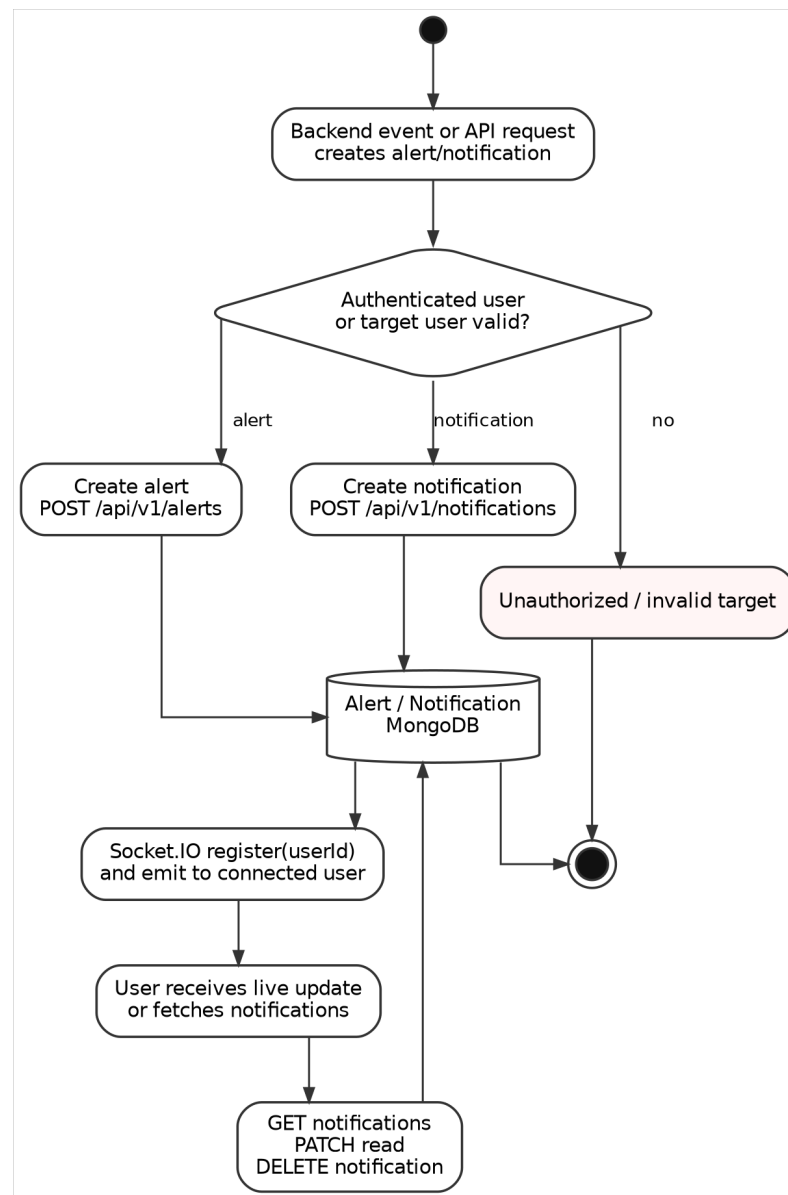
This patient entity workflow documents how a patient record moves through the backend. The patient is created by an admin or caretaker, linked to a caretaker and optionally a nurse or doctor, then accumulates profile updates, health records, reports, logs, prescriptions, and sensor/activity data until the record is eventually deactivated rather than permanently removed.

Organization and Staff Approval Workflow



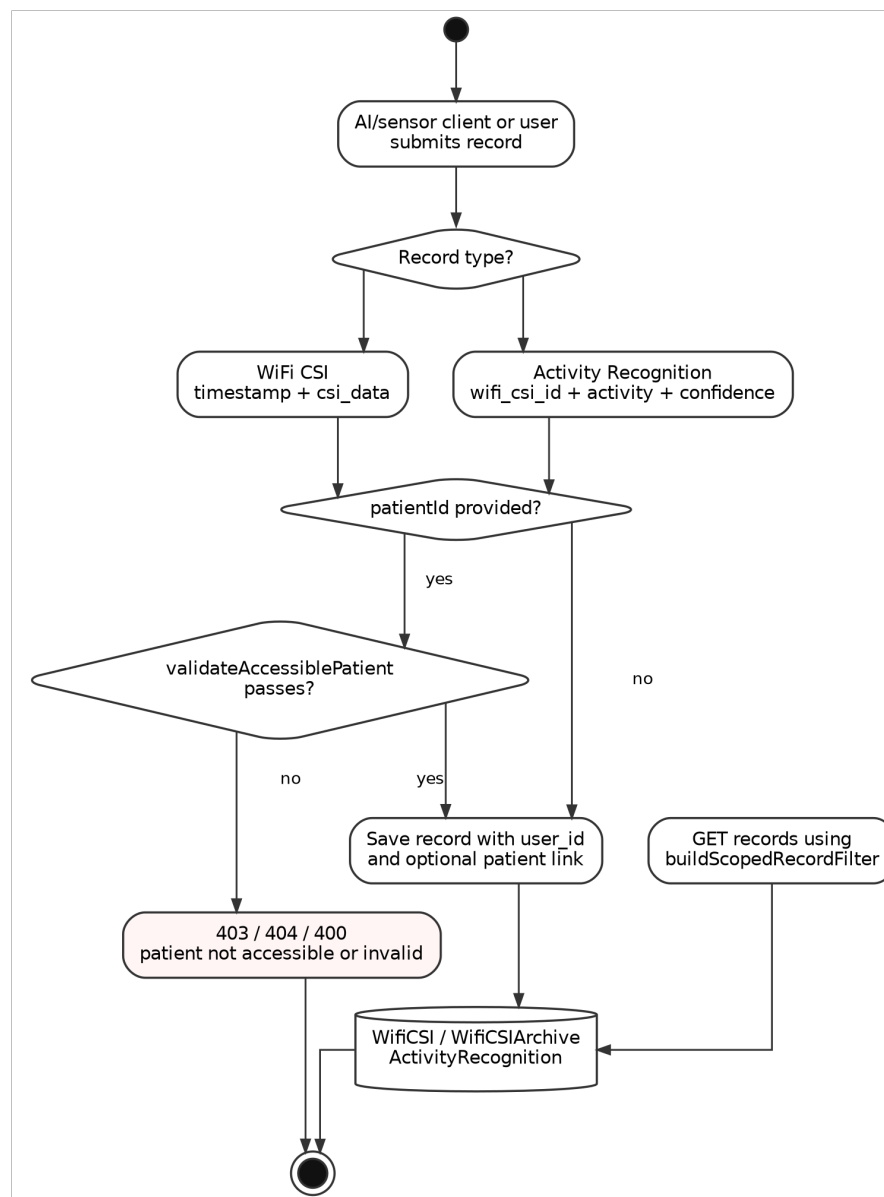
This workflow captures the organization-based access pattern used by the backend. Users register with a role and organization context, the admin links or approves staff for the organization, and the User record stores role, organization, approval status and audit fields so protected organization workflows only continue for valid approved staff.

Alerts and Notifications Workflow - All Users



This workflow shows how alerts and notifications are handled for users. Authenticated requests can create user-specific alerts or notifications, notifications are saved in the database and emitted to connected clients through Socket.IO when the user is registered on the socket connection, and the user can later fetch, mark as read or delete their notifications.

AI and Sensor Data Workflow - WiFi CSI and Activity Recognition



This workflow documents the sensor-related backend path. An authenticated user or integrated client posts WiFi CSI or activity-recognition data, optionally links it to a patientId, the backend validates that the patient is accessible to the authenticated user, and then stores or retrieves records using patient-scoped filters including archived WiFi CSI records.